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Spatiotemporal Modeling of Mature-at-Length Data Using a Sliding Window Approach

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Abstract

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Some key words: Data aggregation; Mature-at-length data; Maturity ogive; Spatiotemporal Model;

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1 Introduction

Assessing maturity status of fish and invertebrate species is important for understanding population dynamics with results potentially influencing fisheries management strategies. For example, maturity information can be used to estimate reproductive potential, such as spawning stock biomass and total egg production, which often serve as inputs to stock assessment models. Growth and maturity information also helps set minimum landing sizes in order to protect juvenile fish. Multiple studies have shown that considerable variability exists in life history traits (e.g. growth rates, maturity schedules) within populations such that intrapopulation spatial variation among regions should not be ignored (Punt et al., 1997; Kai et al., 2017; Olson et al., 2018; Zheng et al., 2020).

Traditionally, the maturity ogive (proportion of matured individuals at length or age) is modelled using a generalized linear model (GLM) with a logit link function (Morgan and Colbourne, 1999). Morgan and Rideout (2008) and Olson et al. (2018) included area factors in the GLM in order to account for the intrapopulation regional variability when modeling size at maturity data for American plaice (*Hippoglossoides platessoides*) and golden king crab (*Lithodes aequispinus*), respectively. Recently, spatiotemporal models have been introduced and widely applied in fisheries science to deal with small-scale spatial variation. Zheng et al. (2020) divided their study area into 50 grids and used a Gaussian Markov random field (GMRF) to model the spatially correlated random effects in their generalized linear mixed model (GLMM) when analyzing maturity data for American plaice. They found the model fit improved with increasing spatial resolution. Even though spatiotemporal models can help reduce sample selection bias and fill spatial gaps (Kai et al., 2017), survey sample coverage may still vary remarkably from year to year due to spatial effort allocation, and can result in biased estimates even with spatiotemporal models.

In this study, we analyze maturity data for Greenland halibut (*Reinhardtius hippoglossoides*) collected by surveys in the eastern Canadian Arctic. Greenland halibut is an important commer-

cial flatfish species in the Arctic that is highly mobile. See Bowering (1984); Boje (2002); Hussey et al. (2017), and Barkley et al. (2018) for studies of their inter-regional migration behavior. Age determination is problematic for Greenland halibut; therefore, we focus on mature-at-length data. The length at which 50% of the stock is mature (L_{50}) is a crucial maturation parameter for the maturity ogive and frequently used in stock assessment. Researchers may sometimes find data from fisheries-independent survey schemes are highly unbalanced, i.e., surveys might be conducted in different regions for different years and there may be missing years. We develop a spatiotemporal GLMM to account for variation in L_{50} through space and time and tackle the sample coverage issue by data aggregation using a sliding window. We fit this spatiotemporal model to each sex separately. In addition, we undertake a simulation study to investigate the estimation bias for L_{50} that exists with conventional models, and compare performance with our sliding window spatiotemporal model.

2 Data

Data are provided by Fisheries and Oceans Canada from stratified-random otter trawl surveys conducted with an Alfredo III trawl in Northwest Atlantic Fisheries Organization (NAFO) Subarea 0 from September to December among years 1999 to 2017. The NAFO Subarea 0 ranges from Davis Strait to Baffin Bay and is further separated into three divisions: 0A-North (N0A), 0A-South (0A), and 0B. See Figure 1 for the spatial range and geographical features of the survey area. For each tow, the total number of Greenland halibut caught was recorded along with other tow specific variables, including start and end locations and dates, tow duration, and average fishing depth. In addition, individual fish were measured for length (rounded to the nearest 1 centimeter), sex, and maturity status. Females with small ovaries and no eggs visible to the naked eye, and males with mostly clear and very small testes, were considered immature. All other maturity stages (maturing levels A, B and C, spawning, and resting) are considered as mature in our analysis. See Morgan and Treble (2006) for more detail on the maturity stages

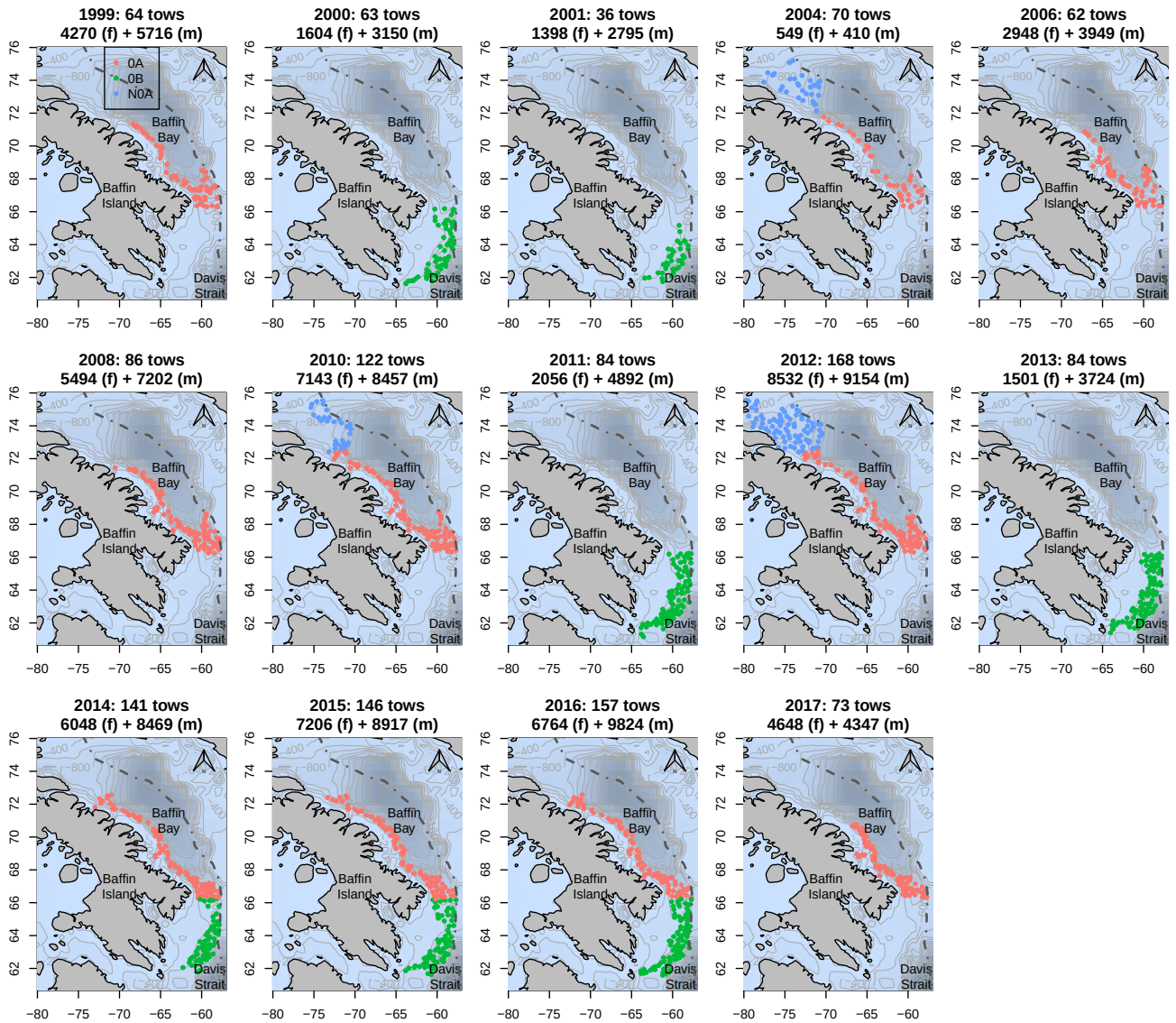


Figure 1: Locations of tows with fully sampled individuals by year, color coded by division (legend shown in the first panel). The number of tows and the number of fully sampled females (f) and males (m) are printed above each panel. The Canadian Exclusive Economic Zone is indicated by a dashed line.

and a first look at Greenland halibut maturity data in this area, Harris et al. (2009) provides an update.

Due to limited resources, surveys could not be conducted yearly for all three divisions, and in some tows, only a subset of Greenland halibut were measured for length, among which a further subset had their sex and maturity status recorded. We hereafter refer to fish with all three attributes available as fully sampled, and those with only length information as partially

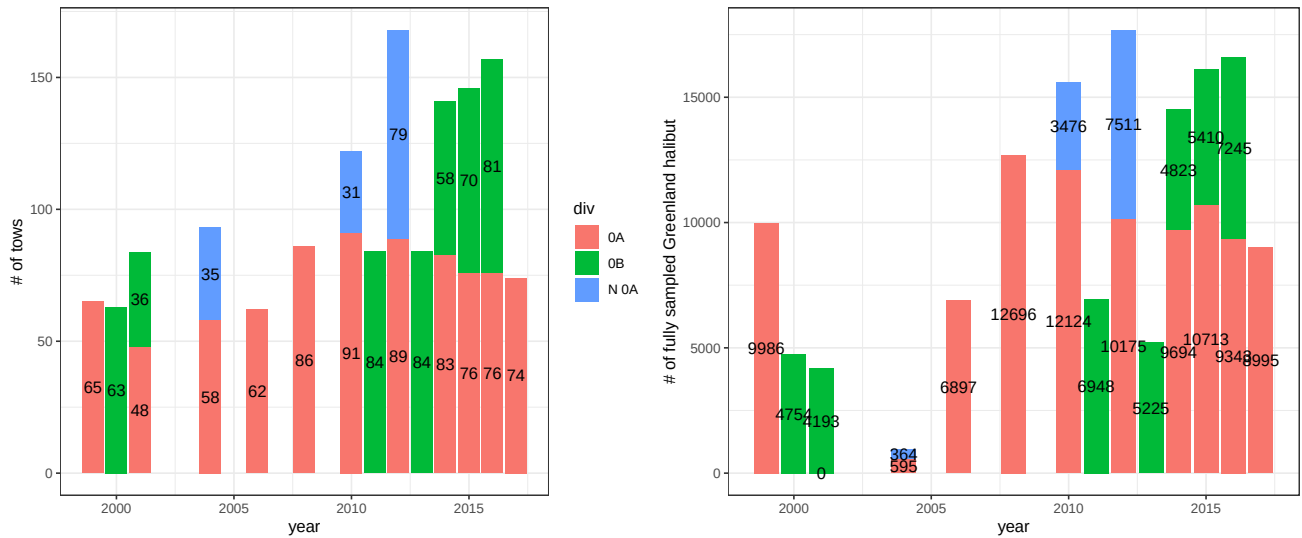


Figure 2: Number of tows (left) and fully sampled individuals (right) by year and division.

sampled. Figure S1 in the supplementary material illustrates the relationship between partially and fully sampled individuals among all the fish counted for each year.

There are in total 1429 tows over 14 years, with 141167 Greenland halibut fully sampled, among which 60161 are females and 81006 are males (Figure 2). In 2001, no individuals were fully sampled in OA. In 2004, only a very small proportion of Greenland halibut were fully sampled in both N OA and OA due to a change in sampling to a length stratified approach (10 fish sample per cm per sex per trip). Figure S2 in the supplementary material presents the length distribution of the fully sampled individuals overlaid for both sexes by year, from which the effect of the change in sampling on the length distribution can be observed. Since 2010, surveys have been conducted annually with an increasing amount of maturity information available, as can be seen by the increasing proportion of fully sampled individuals (in blue) in Figure S1. However, the sample coverage problem still exists. Figure 1 displays the spatial distribution of fully sampled individuals each year, and provides data on the number of tows, and the number of fully sampled females and males. Locations in the three divisions are represented by different colors. We can see the sample coverage is highly unbalanced in the three divisions across years.

3 Spatiotemporal Model for Mature-at-Length Data

Traditionally, for the j th fully sampled individual from the i th tow, the indicator variable M_{ij} for whether the individual is mature (or not), given its sex g_{ij} and length l_{ij} is modelled with a binomial GLM (logistic regression), where the logit of p_{ij} , the probability of being mature, is assumed to be a linear function of length. In order to take spatiotemporal variation in maturity rates into consideration, we add year factors $\beta_{0|G}(t)$ as fixed effects and spatial random effects $\xi_G(\mathbf{s})$ into the model:

$$M_{ij} | (G = g_{ij}, L = l_{ij}) \sim \text{Bernoulli}(p_{ij}), \text{logit}(p_{ij}) = \beta_{1|g_{ij}} \cdot l_{ij} + \beta_{0|g_{ij}}(t_i) + \xi_{g_{ij}}(\mathbf{s}_i), \quad i = 1, \dots, n, \quad j = 1, \dots, n_i, \quad (1)$$

where n is the total number of tows, n_i , t_i and $\mathbf{s}_i$ are the number of fully sampled individuals, year factor and spatial location of the i th tow, respectively. The model parameters are subscripted by sex G (M for male and F for female) to correspond to two sets of parameters for the spatial GLMM (1), one for each sex. The two sets of parameters are estimated separately. We model $\xi_G(\mathbf{s})$ as a mean zero Gaussian random field (GRF) with a Matérn covariance function. The Matérn covariance function is isotropic such that the correlation between $\xi_G(\mathbf{s}_i)$ and $\xi_G(\mathbf{s}_j)$ depends only on the distance between the two locations. The Matérn covariance function is controlled by variance parameter σ_G^2 , range parameter ϕ_G that determines how fast the spatial correlation decreases with increasing distance, and a smoothness parameter ν_G that governs the mean square differentiability of the spatial process $\xi_G(\mathbf{s})$. Then $L_{50|G}$ can be derived from model (1) by setting $p = 0.5$ upon which it can be written as a ratio of the spatial-temporally varying intercept $\beta_{0|G}(t) + \xi_G(\mathbf{s})$ and the slope $\beta_{1|G}$:

$$L_{50|G}(\mathbf{s}, t) = -\frac{\beta_{0|G}(t) + \xi_G(\mathbf{s})}{\beta_{1|G}}. \quad (2)$$

The spatial-temporally varying $L_{50|G}(\mathbf{s}, t)$ can be decomposed into a consistent spatial pattern (by fixing $\beta_{0|G}(t)$ at the value of a given year) and a temporal trend (by setting $\xi_G(\mathbf{s}) = 0$), additively.

Instead of using a linear function of length on the logit scale of p as is the conventional approach, Zheng et al. (2020) model $\log(L_{50|G})$ and log of the difference between L_{95} (length at 95% maturity) and L_{50} directly as a linear function of length with spatiotemporal random effects with an autoregressive structure in time for both $\log(L_{50})$ and log of the difference. Their model has the advantage that the constraints $L_{50} > 0$ and $L_{95} > L_{50}$ are incorporated implicitly. Their model is comparable to an extension of (1) with spatiotemporal random effects added to both the intercept and slope, which result in changes in both shifting (maturing at smaller or bigger size) and tilting (maturing faster or slower) of the maturity ogive. Including both a spatiotemporally varying intercept and slope lead to a more flexible model, but at the cost of high variability in parameter estimation. Greenland halibut have a long life span, up to 33 years (Treble et al., 2008), compared to 24 years for American plaice (Zheng et al., 2020) and our study spans only 19 years compared to the 38 years available to Zheng et al. (2020); therefore, for Greenland halibut it is unreasonable to include both a spatiotemporally varying intercept and slope. A simplified version of the model in Zheng et al. (2020) with only a spatial-temporally varying L_{50} can be regarded as a reparameterization of (1) and will essentially yield the same results. Therefore, we continue with our formulation of model (1).

4 Data Aggregation Using a Sliding Window

By taking the intra-population small-scale spatial variation of maturity schedule into account in model (1), we aim to get better estimation of $L_{50|G}$ and its temporal trend. However, if sample coverage is too unbalanced in space across years, problems can still arise. To deal with this sample coverage problem, we aggregate tows using a sliding window of 5 years. That is, the aggregated data for the k th sliding window consist of all data obtained during years $1998 + k$ to $2003 + k$, for $k = 1, \dots, 15$. Figure 3 presents the aggregated data for the 16 sliding-window periods. We can see the sample coverage is much more consistent across sliding-window years than across years for the original data shown in Figure 1. The sliding window size (i.e. 5 years)

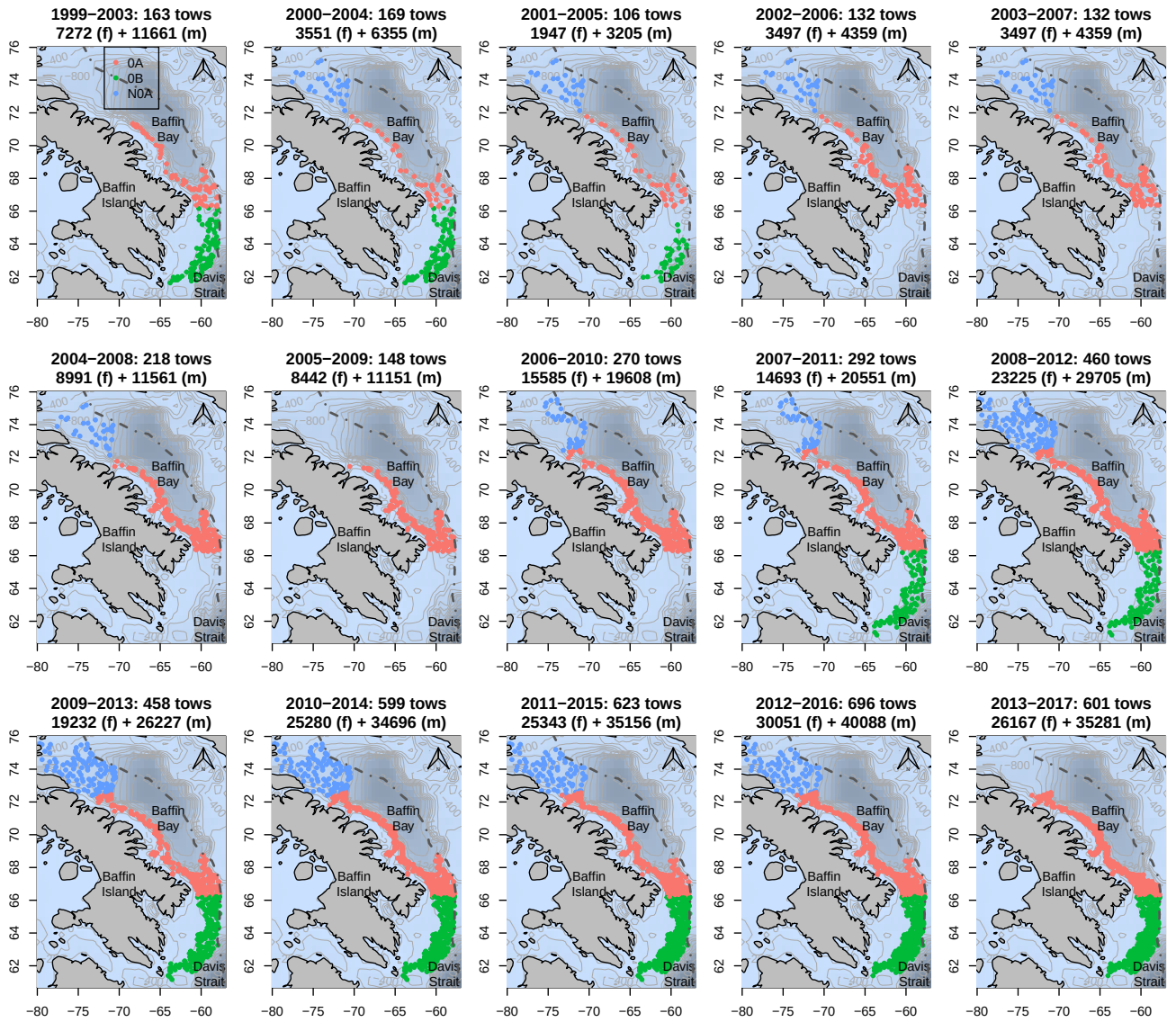


Figure 3: Locations of aggregated tow data with fully sampled individuals by sliding window and division, colored according to the corresponding division and labelled with the number of tows, fully sampled females (f) and males (m) for each sliding window. The Canadian Exclusive Economic Zone is shown by dashed lines.

is chosen by considering a trade-off between balanced sample coverage and temporal resolution of changes in maturity. With a sliding window's width smaller than 5 years, there will still be sample coverage problems for some sliding-window periods; while with width larger than 5 years we may fail to capture a possible temporal trend in maturity. Note that, with the 5-year sliding window, the aggregated data for the 4th and 5th sliding-window periods are exactly the same, and the 7th sliding window data only involve tows from OA.

We fit the spatial GLMM in (1) with both the original and sliding window data, for each sex separately. Estimation of the GRF is computationally expensive, since it involves inversions of $n \times n$ matrices, which becomes infeasible for thousands of tow locations. We approximate the GRF continuously indexed in space by a GMRF at discrete locations with a sparse precision matrix (the inverse of a covariance matrix) via the stochastic partial differential equations (SPDE) approach (Lindgren et al., 2011). First, a mesh is created by triangulation of the spatial domain with knots placed near the tow locations along with some additional locations such that the maximum edge length does not exceed a prescribed value that controls the spatial resolution, and that the minimum edge length is greater than a cut-off value for computational efficiency. The R-INLA package (Rue et al., 2009) provides useful tools for creating desirable meshes for practitioners. For parameter estimation, we implement model (1) using the Template Model Builder (TMB) package (Kristensen et al., 2016) in the R software (R Core Team, 2021). TMB computes integrals involved in the likelihood of a GLMM via the Laplace approximation and utilizes automatic differentiation (Griewank and Walther, 2008) to assist optimization. Then, maximum likelihood estimates (MLE) of the model parameters can be obtained and the spatial random effects $\xi_G(\mathbf{s})$ can be interpolated at any given location. Then the spatiotemporally varying $L_{50|G}(\mathbf{s}, t)$ can be derived.

5 Simulation Study

We undertake a simulation study to demonstrate estimation problems with L_{50} when using conventional models caused by unbalanced sampling. We compare performance with our spatiotemporal model using sliding window data.

We simulate maturity data from model (1) using covariate information (length, year, division, spatial location) from the 60161 female Greenland halibut individuals, with $\beta_1 = 0.2$, an increasing $\beta_0(t)$ from -16 to -14 over years (this parameter setting results in a linearly decreasing L_{50} from 80cm to 70cm for the period 1999 to 2017) and spatial random fields $\xi(\mathbf{s})$ estimated

from the sliding window data for females. For each simulation, we fit three models: a binomial GLM with length and year factors, a binomial GLM with length, year and division factors (0A as reference level, hereafter referred to as the divisional GLM), and our spatial GLMM (1), with both the original and sliding window data. For each model fit, from the estimated parameters we derive the yearly varying L_{50} by $\hat{L}_{50}(t) = -\frac{\hat{\beta}_0(t)}{\hat{\beta}_1}$ at the surveyed years. Results for the estimated $\hat{L}_{50}(t)$ curve from 100 replicates are shown by functional boxplots (Sun and Genton, 2011) in Figure 4. Functional boxplot treats each $\hat{L}_{50}(t)$ curve as a whole instead of having to compute an individual boxplot of $\hat{L}_{50}(t_i)$ at each surveyed year. Similar to a boxplot, the median curve, 50% central region (counterpart of the box), maximum non-outlying envelope (counterpart of the whisker), and outlying curves are indicated by a bold black line, magenta area, white area, and red dashed line, respectively. The true L_{50} used to generate the data is indicated by a green line in each functional boxplot.

With the original data, the $L_{50}(t)$ estimated by the GLM zigzags between years even though the true L_{50} decreases linearly. The GLM does not take the intrapopulation spatial variation in L_{50} into account and the estimated $L_{50}(t)$ is greatly affected by the area surveyed each year. This is most obvious for years 2010 to 2013, when the survey area alternated between 0A and 0B. The divisional GLM mitigates the zigzag effect (for years 2010 to 2013) and reduces overall bias in L_{50} by adding division factors. With fixed division effects and their coarse spatial resolution, the divisional GLM still cannot capture well the true trend in L_{50} . The spatial GLMM improves estimation of $L_{50}(t)$ remarkably by including spatially correlated random effects that can borrow information from nearby locations. However, the overall L_{50} is underestimated (median curve in black is below the truth in green). With the sliding window data, all three methods are less affected by the unbalanced sampling and can better capture the declining trend in L_{50} , while our spatial model performs best. With the sliding window data, our spatial model also corrects bias in estimation of L_{50} .

Maturity data simulated using covariates of the male Greenland halibut (with spatial random

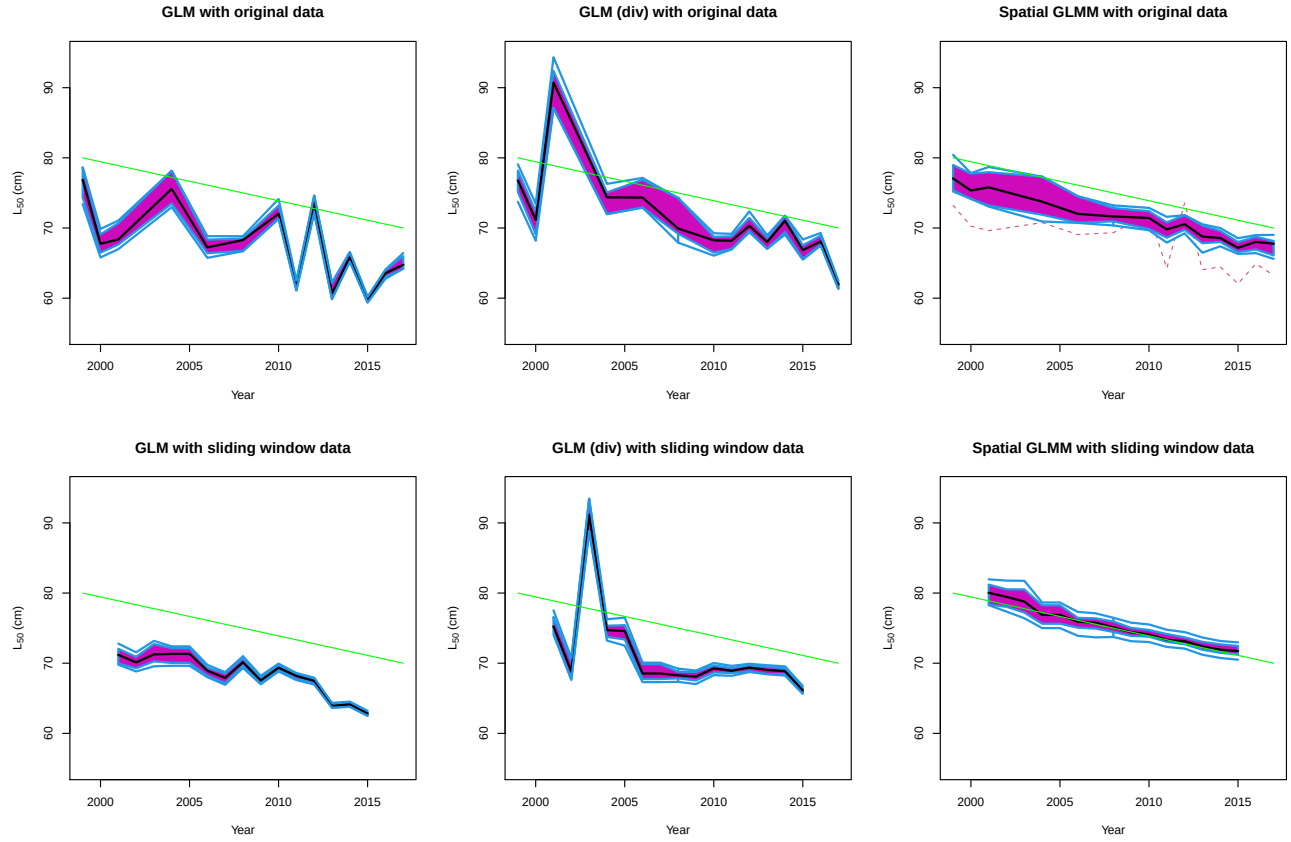


Figure 4: Functional boxplots of the estimated L_{50} from 100 simulations (maturity data generated using covariates of the female fish) by three methods: GLM, GLM with division factors, and spatial GLMM with the original data (upper row) and sliding window data (lower row). The true L_{50} with a decreasing trend is indicated in green line.

fields $\xi(\mathbf{s})$ estimated from the sliding window data for males, $\beta_1 = 0.2$ and a linearly increasing $\beta_0(t)$ from -16 to -14) show similar results (Figure 5). The variation in 2004 in the GLM model with original data is more pronounced and could be linked to the change in sampling protocol that year. Also, the divisional GLM has high peaks in both 2001 and 2008, and there is an outlying curve for the spatial GLMM that was not seen for females. However, the general pattern in the simulations hold and the spatial model with sliding window data improves the estimation of L_{50} for males as well as females.

The simulation study shows first that estimation of yearly patterns of L_{50} is heavily affected by the regions sampled when using a method that does not take intrapopulation variation in maturity rates into account. Second, estimation by the spatial model with the original data

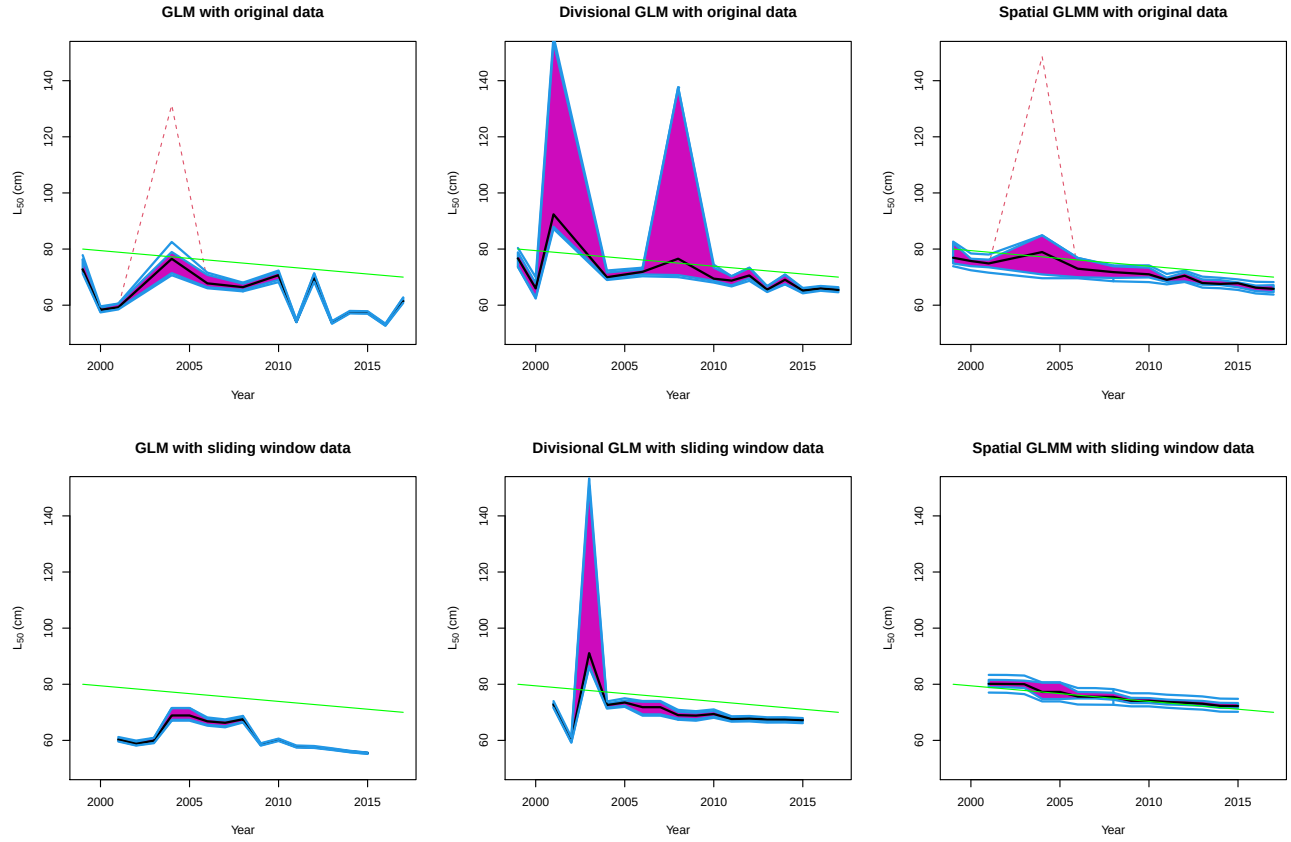


Figure 5: Functional boxplots of the estimated L_{50} from 100 simulations (maturity data generated using covariates of the male fish) by three methods: GLM, GLM with division factors, and spatial GLMM with the original data (upper row) and sliding window data (lower row). The true L_{50} with a decreasing trend is indicated in green line.

can capture the L_{50} trend better, however, there still remains some bias. Culminating, with the sliding window data, estimation of L_{50} is less affected by unbalanced sampling and changes in sampling protocol for all three methods, with the spatial model fully capturing the true trend and without bias.

6 Results

Figure 6 exhibits the spatial variation in the L_{50} at the reference year 1999 (or the first sliding window period) and accompanying standard errors, while Figure 7 presents changes in L_{50} over time, for both females and males and with both original and sliding window data. In Figure 7, the temporal trend of L_{50} estimated from the sliding window data is plotted at the mid-point

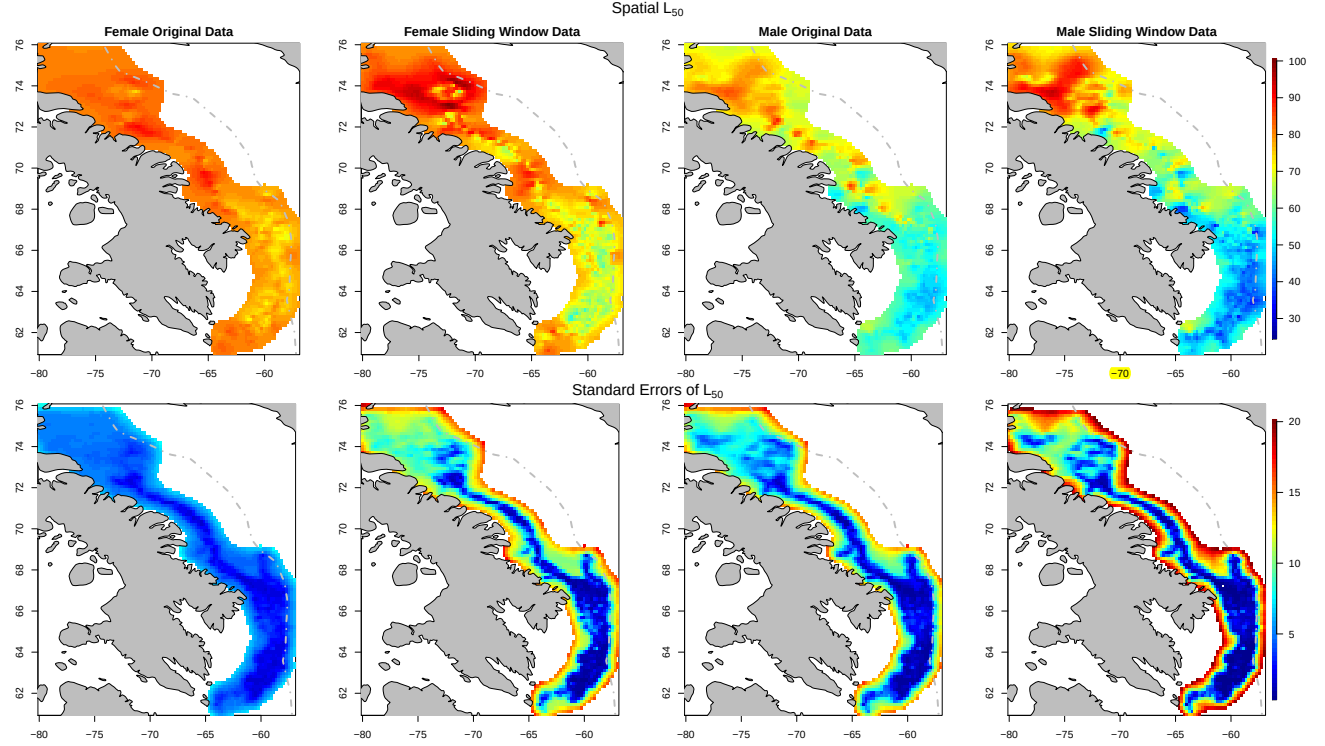


Figure 6: Spatial variation of L_{50} (upper row) at the reference year 1999 or the first sliding window period and the corresponding standard errors (lower row), for both females and males and estimated from both original and sliding window data.

The spatial pattern in L_{50} shown in the first row of Figure 6 demonstrates that L_{50} is much larger in the north for both sexes. For females, $L_{50|F}(\mathbf{s})$ estimated from the sliding window data is on average 70cm in the south (of 68 N°) and 82cm in the north; while for male, $L_{50|M}(\mathbf{s})$ ranges from 50cm in the south to 72cm in the north. This conforms with previous studies in maturity assessment of Greenland halibut (Harris et al., 2009; Kennedy et al., 2014). The simulation study produced similar results, L_{50} estimated with the original data is smaller overall and has a smaller spatial variability than L_{50} obtained with the sliding window data. Fitting our spatial GLMM with the original data may lead to an underestimation of the true L_{50} and its variation in space as demonstrated by our simulation study (Figure 4). Figure 6 also illustrates that the

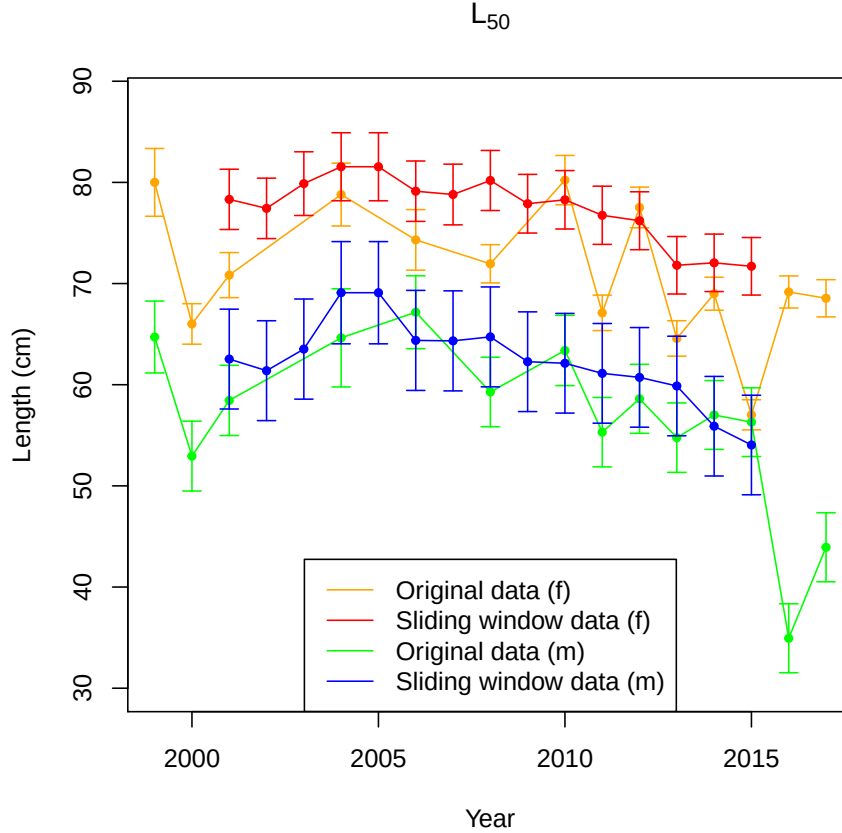


Figure 7: Plots of mean L_{50} in time for both females and males and with both original or sliding window data.

The temporal trend in L_{50} estimated for the stock using original data oscillates substantially among years, while L_{50} estimated using sliding window data is smoother over time (Figure 7).

The model using sliding window data indicates a change in L_{50} for female Greenland halibut from approximately 80 cm to 73 cm, and for males from about 65 cm to 55 cm. Figure 6 and 7 also show that in general female Greenland halibut mature at a much bigger size than males,

which is naturally linked to their sexual dimorphism, with females growing to a larger size than males.

7 Discussion

While it is pleasing to see spatiotemporal models becoming increasingly popular and used by scientists from different fields, it is important that users be aware that issues may arise due to highly unbalanced sample coverage over space and/or time. In our fishery application, our simulation study demonstrates that applying spatial GLMMs to mature-at-length data with highly unbalanced sample coverage may produce biased estimates of L_{50} . We address this sample coverage issue by performing data aggregation using a sliding window. Although our focus was on data from an Arctic marine fish survey and length-at-maturity data, the sliding window data aggregation approach could be applied to other types of data where sample coverage issues exist. For example, surveys, biotelemetry, and sample collection programs in Arctic marine ecosystems are often impacted by logistical challenges and high costs (Mallory et al., 2018) that can result in infrequent sampling (Yurkowski et al., 2019, Table 1). In particular, aerial surveys of marine mammal abundance are conducted in different areas and years, and recently they have been expanding to areas not previously surveyed (Biddlecombe and Watt, 2021). There are also spatial and temporal gaps in other types of data (e.g. fatty acids and stable isotopes) used to examine food-web trends in marine ecosystems that can lead researchers to group data into broad categories to try to address sampling gaps (Watt and Ferguson, 2015). Further, in surveys of bird populations, there are problems with coverage both in terms of area and years. For instance in Kejimikujik National park in Nova Scotia their Forest Bird Species Population and Biodiversity Monitoring has had gaps in coverage both in years and point counts (personal communication Dr. Cindy Staicer). Also, the Birds Canada study on High Elevation Landbird Program (Kouwenberg, 2019) which has been monitoring the status of the threatened Bicknell’s thrush has had a number of changes in transects and point count locations since 2002

as well as changes in detection methods. Finally, in epidemiology, data sparseness has been reported when modelling air pollution (e.g. Paciorek et al. (2009); Rushworth et al. (2014)) or the behaviour of infectious diseases (e.g. Hampton et al. (2011); Rumisha et al. (2014)). While conceptually straightforward, the sliding window approach shows much promise for a variety of settings mentioned above.

In this paper, we fitted our spatial GLMM to mature-at-length data of Greenland halibut from surveys in the eastern Canadian Arctic for both sexes, and with both the original and sliding window data. The spatial pattern in L_{50} shows it to be much larger in Baffin Bay compared to Davis Strait for both males and females. This observed difference in L_{50} between the north and south may be due to a combination of factors related to migration and the maturation process for this species. For example, research indicates a majority of mature fish migrate to spawning grounds in the Davis Strait, including Division 0B (Templeman, 1973; Jørgensen, 1997) and Greenland halibut appear to have a prolonged adolescence and/or a multi-year maturation process (Simonsen and Gundersen, 2005; Kennedy et al., 2011; Rideout et al., 2012). A relatively few large mature fish amongst a group of immature or non-spawning fish of various size could result in a shift to a higher L_{50} in northern feeding areas. Fitting results using sliding window data reveal a declining trend in L_{50} in recent years for both sexes. This declining trend might be related to increasing fishing pressure that has been demonstrated to lead to a similar response, younger and smaller sizes at maturity, for other fish species (Morgan and Colbourne, 1999; Wright et al., 2011; Zheng et al., 2020). Whether the hypothesis that the declining trend is related to increasing fishing pressure and decreasing abundance of Greenland halibut is an interesting topic for exploration.

In this study, length-at-mature data for females and males were modelled separately. Extensions could be made by including an inter-sexual correlation and fitting the full data simultaneously as did Zheng et al. (2020). Moreover, it can be difficult to determine maturity status for males and errors may arise based on the subjective description of the macroscopic maturity

scale. Kennedy et al. (2014) compared L_{50} for both the macroscopic (visual) scale and the more accurate microscopic (histological) scale for maturity of Greenland halibut and found an increase in L_{50} for the microscopic maturity data in East Greenland. Another useful extension might be to incorporate potential length measurement errors into the spatial GLMM.

A next step in this study is to expand the spatial domain to a much larger scale by combining data from surveys in the gulf of St. Lawrence, and Newfoundland and Labrador shelves in order to examine spatiotemporal patterns in maturity for signs of population or stock differences.

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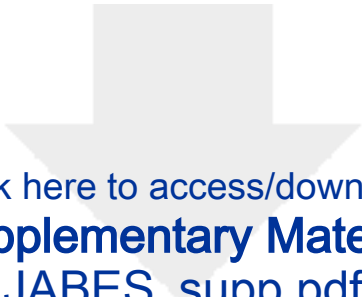
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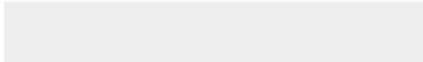
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